

Emerging and re-emerging swine viruses.



In the past two decades or so, a number of viruses have emerged in the global swine population. Some, such as porcine reproductive and respiratory syndrome virus (PRRSV) and porcine circovirus type 2 (PCV2), cause economically important diseases in pigs, whereas others such as porcine torque teno virus (TTV), now known as Torque teno sus virus (TTSuV), porcine bocavirus (PBoV) and related novel parvoviruses, porcine kobuvirus, porcine toroviruses (PToV) and porcine lymphotropic herpesviruses (PLHV), are mostly subclinical in swine herds. Although some emerging swine viruses such as swine hepatitis E virus (swine HEV), porcine endogenous retrovirus (PERV) and porcine sapovirus (porcine SaV) may have a limited clinical implication in swine health, they do pose a potential public health concern in humans due to zoonotic (swine HEV) or potential zoonotic (porcine SaV) and xenozoonotic (PERV, PLHV) risks. Other emerging viruses such as Nipah virus, Bungowannah virus and Menangle virus not only cause diseases in pigs but some also pose important zoonotic threat to humans. This article focuses on emerging and re-emerging swine viruses that have a limited or uncertain clinical and economic impact on pig health. The transmission, epidemiology and pathogenic potential of these viruses are discussed. In addition, the two economically important emerging viruses, PRRSV and PCV2, are also briefly discussed to identify important knowledge gaps. © 2012 Blackwell Verlag GmbH.